TOPIC 2 Transforming and Comparing Linear Functions

Name _____ Date _____

Reminders

Slope = $(y_2 - y_1) \div (x_2 - x_1)$ · $y = mx + b$ · point-slope: $y - y_1 = m(x - x_1)$

Parallel: same slope · Perpendicular: flip the fraction and change the sign (product −1)

Outside f(x) + k: up/down · Inside f(x − h): right h, f(x + h): left h (opposite the sign)

Horizontal: y = #, slope 0 · Vertical: x = #, slope undefined (HOY VUX)

Domain = inputs (x) · Range = outputs (y) · Zero = x-intercept

1. The graph of linear function m passes through $(-4, 6)$ and $(2, -3)$. Given $f(x) = x$ and $m(x) = af(x)$, what is the value of a?

2. A food truck rents for $120 plus $45 per hour, for at most 6 hours. What is the range for one rental?

F. $0 \le x \le 6$

G. $45 \le y \le 390$

H. $120 \le y \le 270$

J. $120 \le y \le 390$

3. A vertical line passes through $(8, -3)$. What are its equation and slope?

A. $x = 8$; slope is undefined

B. $y = -3$; slope is zero

C. $x = 8$; slope is zero

D. $y = 8$; slope is undefined

4. Lena graphed $f(x)$ and $g(x) = f(x + 4)$. Which statement describes how the graphs are related?

F. translated right 4 units

G. translated left 4 units

H. translated up 4 units

J. translated down 4 units

5. $f(x)$ has the table $(-1, 8)$, $(0, 5)$, $(1, 2)$, $(2, -1)$. The graph is translated 3 units left. What is the y-intercept of the new function?

6. $f(x) = x$ and $g(x) = f(\frac{1}{2}x)$. Write the equation for $g(x)$.

7. Write the equation in slope-intercept form of the line that crosses the x-axis at 8 and is perpendicular to $y = 4x + 1$.

8. Write the equation in slope-intercept form of the line through (−6, 0) that is parallel to $y = 1.5x - 4$.

9. A line passes through (1, −2) and (4, 7). Write an equation of the line in point-slope form.

10. A line passes through (0, 6) and (3, 0). Which statement is true?

F. It is decreasing and its zero is 6.

G. It is increasing and its zero is 3.

H. Its slope is $-1/2$ and its zero is 3.

J. It is decreasing and its zero is 3.

11. Which pair of equations represents perpendicular lines?

A. $y = 2/3x - 1$ and $y = -3/2x + 4$

B. $y = 2/3x + 1$ and $y = 2/3x - 4$

C. $y = 2/3x$ and $y = 3/2x + 2$

D. $y = -2/3x + 1$ and $y = -3/2x$

12. A phone is at 80% battery and loses 4% per hour until it is empty. What is the domain?

F. $0 \le x \le 80$

G. $0 \le x \le 20$

H. $0 \le y \le 80$

J. $4 \le x \le 80$